MSG No. 136.505 P_C4_2/mnm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `m.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,6,9,14,23,24,31,32]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,10,11,20,21,28,29]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,5,12,13,18,19,26,27]`
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[7,8,15,16,17,22,25,30]`

Table 2: Wyckoff site: `4b`, site symmetry: `m.mm`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,6,7,8,9,14,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,4,5,10,11,12,13]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[17,22,23,24,25,30,31,32]`
4	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[18,19,20,21,26,27,28,29]`

Table 3: Wyckoff site: `4c`, site symmetry: `m.m'm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[1,7,14,16,22,24,25,31]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[2,5,11,12,19,20,26,29]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[3,4,10,13,18,21,27,28]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[6,8,9,15,17,23,30,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `m.m'm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[1,7,14,16,22,24,25,31]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[2,5,11,12,19,20,26,29]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[3,4,10,13,18,21,27,28]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[6,8,9,15,17,23,30,32]`

Table 5: Wyckoff site: `4e`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6,10,11,20,21,31,32]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,9,14,23,24,28,29]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[4,5,15,16,17,22,26,27]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7,8,12,13,18,19,25,30]`

Table 6: Wyckoff site: `4f`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,6,15,16,20,21,26,27]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,12,13,23,24,25,30]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[4,5,10,11,17,22,31,32]`
4	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7,8,9,14,18,19,28,29]`

Table 7: Wyckoff site: `8g`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,31,32]`
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3,28,29]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[4,5,26,27]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8,25,30]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,14,23,24]`
6	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,11,20,21]`
7	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[12,13,18,19]`
8	$\left[\frac{1}{2},\ 0,\ z\right]$	`[15,16,17,22]`

Table 8: Wyckoff site: `8h`, site symmetry: `2.mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3,12,13]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,5,10,11]`
4	$[0,\ 0,\ -z]$	`[7,8,9,14]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[17,22,31,32]`
6	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[18,19,28,29]`
7	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[20,21,26,27]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[23,24,25,30]`

Table 9: Wyckoff site: `8i`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[1,16,22,31]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[2,12,19,29]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[3,13,18,28]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[4,10,21,27]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[5,11,20,26]`
6	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[6,15,17,32]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[7,14,24,25]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[8,9,23,30]`

Table 10: Wyckoff site: `8j`, site symmetry: `m.2'm'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	`[1,14,23,32]`
2	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[2,11,21,28]`
3	$\left[x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[3,10,20,29]`
4	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[4,13,19,26]`
5	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	`[5,12,18,27]`
6	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	`[6,9,24,31]`
7	$\left[x+\frac{1}{2},\, x,\, 0\right]$	`[7,16,17,30]`
8	$\left[\frac{1}{2}-x,\, -x,\, 0\right]$	`[8,15,22,25]`

Table 11: Wyckoff site: `8k`, site symmetry: `m.2m`

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{2}\right]$	`[1,7,14,16]`
2	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, 0\right]$	`[2,5,11,12]`
3	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	`[3,4,10,13]`
4	$\left[-x,\, -x,\, \frac{1}{2}\right]$	`[6,8,9,15]`
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[17,23,30,32]`
6	$[-x,\, x,\, 0]$	`[18,21,27,28]`
7	$[x,\, -x,\, 0]$	`[19,20,26,29]`
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[22,24,25,31]`

Table 12: Wyckoff site: `8l`, site symmetry: `m.2'm'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[1,14,23,32]`
2	$\left[-x,\, x+\frac{1}{2},\, 0\right]$	`[2,11,21,28]`
3	$\left[x,\, \frac{1}{2}-x,\, 0\right]$	`[3,10,20,29]`
4	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	`[4,13,19,26]`
5	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	`[5,12,18,27]`
6	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[6,9,24,31]`
7	$\left[x+\frac{1}{2},\, x,\, \frac{1}{2}\right]$	`[7,16,17,30]`
8	$\left[\frac{1}{2}-x,\, -x,\, \frac{1}{2}\right]$	`[8,15,22,25]`

Table 13: Wyckoff site: `8m`, site symmetry: `m.2m`

No.	position	mapping
1	$[x,\, x,\, 0]$	`[1,7,14,16]`
2	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[2,5,11,12]`
3	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[3,4,10,13]`
4	$[-x,\, -x,\, 0]$	`[6,8,9,15]`
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	`[17,23,30,32]`
6	$\left[-x,\, x,\, \frac{1}{2}\right]$	`[18,21,27,28]`
7	$\left[x,\, -x,\, \frac{1}{2}\right]$	`[19,20,26,29]`

continued ...

Table 13

No.	position	mapping
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	[22,24,25,31]

Table 14: Wyckoff site: 16n, site symmetry: .2'.

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	[1,21]
2	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{3}{4}\right]$	[2,24]
3	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[3,23]
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[4,22]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[5,17]
6	$\left[0,\ -y,\ \frac{1}{4}\right]$	[6,20]
7	$\left[y,\ 0,\ \frac{3}{4}\right]$	[7,19]
8	$\left[-y,\ 0,\ \frac{3}{4}\right]$	[8,18]
9	$\left[0,\ -y,\ \frac{3}{4}\right]$	[9,29]
10	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[10,32]
11	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{4}\right]$	[11,31]
12	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[12,30]
13	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[13,25]
14	$\left[0,\ y,\ \frac{3}{4}\right]$	[14,28]
15	$\left[-y,\ 0,\ \frac{1}{4}\right]$	[15,27]
16	$\left[y,\ 0,\ \frac{1}{4}\right]$	[16,26]

Table 15: Wyckoff site: 16o, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\ \frac{1}{2}-x,\ z\right]$	[1,31]
2	$\left[x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[2,29]
3	$\left[-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3,28]
4	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[4,27]
5	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[5,26]
6	$\left[-x,\ x+\frac{1}{2},\ z\right]$	[6,32]
7	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[7,25]
8	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[8,30]
9	$\left[-x,\ x+\frac{1}{2},\ -z\right]$	[9,23]
10	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[10,21]
11	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11,20]
12	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	[12,19]
13	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[13,18]
14	$\left[x,\ \frac{1}{2}-x,\ -z\right]$	[14,24]
15	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[15,17]
16	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[16,22]

Table 16: Wyckoff site: `16p`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ -x,\ z]$	`[1,15]`
2	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,13]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3,12]`
4	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,11]`
5	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[5,10]`
6	$[-x,\ x,\ z]$	`[6,16]`
7	$[-x,\ x,\ -z]$	`[7,9]`
8	$[x,\ -x,\ -z]$	`[8,14]`
9	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	`[17,31]`
10	$\left[x,\ x,\ z+\frac{1}{2}\right]$	`[18,29]`
11	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	`[19,28]`
12	$\left[x,\ x,\ \frac{1}{2}-z\right]$	`[20,27]`
13	$\left[-x,\ -x,\ \frac{1}{2}-z\right]$	`[21,26]`
14	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	`[22,32]`
15	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -z\right]$	`[23,25]`
16	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	`[24,30]`

Table 17: Wyckoff site: `16q`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14]`
2	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,11]`
3	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,10]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[4,13]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,12]`
6	$[-x,\ -y,\ 0]$	`[6,9]`
7	$[y,\ x,\ 0]$	`[7,16]`
8	$[-y,\ -x,\ 0]$	`[8,15]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[17,30]`
10	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[18,27]`
11	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[19,26]`
12	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[20,29]`
13	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[21,28]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ 0\right]$	`[22,25]`
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[23,32]`
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	`[24,31]`

Table 18: Wyckoff site: `32r`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[4]`

continued ...

Table 18

No.	position	mapping
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[5]
6	$[-x,\, -y,\, z]$	[6]
7	$[y,\, x,\, -z]$	[7]
8	$[-y,\, -x,\, -z]$	[8]
9	$[-x,\, -y,\, -z]$	[9]
10	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$[-y,\, -x,\, z]$	[15]
16	$[y,\, x,\, z]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[17]
18	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[18]
19	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[19]
20	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[22]
23	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[23]
24	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[25]
26	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[26]
27	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[27]
28	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[29]
30	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[32]